

Decision-Focused Demand Learning for Equitable Bus Frequency Optimization

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Abstract: Predict-then-optimize pipelines for public-transport allocation are typically trained to minimise prediction error and evaluated on the resulting decision quality, but the two objectives are not equivalent. We study this dissociation on the bus frequency allocation problem of Pune’s PMPML network: 340 routes, five operating periods, and a fleet of approximately 1,800 buses. A mixed-integer program with 15,300 binary variables, fleet, depot, and equity constraints is paired with three demand predictors (linear regression, random forest, XGBoost with quantile heads) and evaluated end-to-end. Decision-focused training is implemented via a sample-weighted L1 surrogate that approximates SPO+ at fixed reference allocation. The decision-focused neural network attains a 44.7% lower decision regret than the MSE-trained XGBoost baseline despite a 13.5% larger test RMSE, demonstrating that prediction quality and decision quality dissociate cleanly on this problem. Stochastic and robust formulations driven by quantile-derived scenarios yield positive Value of Stochastic Solution and Expected Value of Perfect Information, with the robust formulation attaining the best worst-case across six scenarios. Sensitivity analyses characterise the marginal value of fleet expansion (1,500–2,000 passenger-hours per 200 buses) and locate the equity-feasibility cliff at four buses per hour for priority routes; the current floor of three buses per hour costs 0.9% of total waiting time. Against current PMPML frequencies, the optimum reduces total waiting time by 26.0% at the same fleet size.

1 INTRODUCTION

Public bus systems in Indian metropolitan areas operate under a combination of constraints that few transit planning tools were designed to address jointly: heavy-tailed demand, fleet sizes that have not kept pace with population growth, and equity considerations that bind on both political and welfare grounds. The Pune Mahanagar Parivahan Mahamandal Limited (PMPML), which serves over 800,000 daily riders across 340 routes with a fleet of approximately 1,800 buses, is representative. Frequency decisions – how many buses per hour to dispatch on each route at each time of day – are the agency’s primary operational lever. They are currently chosen by heuristic rules of thumb informed by historical practice and disaggregated demand reports.

This paper formulates frequency allocation as a mixed-integer program (MIP), trains a demand model to drive it under uncertainty, and connects the predictor to the optimiser via a decision-focused training procedure that targets operational regret rather than

prediction error. The combination is motivated by an empirical phenomenon that we document quantitatively in §4: prediction quality and decision quality dissociate. A predictor that is uniformly worse on standard forecasting metrics (RMSE, MAPE) can produce decisions that are uniformly better on the downstream cost. The conventional approach of selecting predictors by held-out RMSE is therefore not optimal for predict-then-optimize pipelines, and the gap is large enough to be operationally significant rather than statistical noise.

We contribute (i) a deterministic frequency-allocation MIP with explicit fleet, depot, and equity constraints that scales to the full PMPML network and solves to provable optimality in seconds; (ii) an XGBoost demand-forecasting pipeline with quantile heads that supplies expected demand and uncertainty bands; (iii) stochastic and robust (minimax) MIP formulations driven by the quantile-derived scenarios, with explicit decomposition of the Value of Stochastic Solution (VSS) and Expected Value of Perfect Information (EVPI); (iv) a decision-focused neural

network trained with an SPO+-style surrogate that achieves 44.7% lower regret than the MSE-trained XGBoost despite worse RMSE; and (v) a sensitivity study over fleet size and equity threshold, with LP shadow-price analysis identifying the binding depot bottlenecks. Code, data, and reproduction scripts will be released upon acceptance.

2 RELATED WORK

The classical literature on bus frequency setting begins with (Furth and Wilson, 1981) and (Ceder and Wilson, 1986), who established the trade-off between rider waiting cost and operator capacity cost and provided heuristics that remain in use at many transit agencies. Our work updates two pieces of this pipeline: where the classical literature treats demand as exogenous, we predict it with modern machine learning; where the classical optimisation is single-scenario, we explicitly handle uncertainty.

The seam between learning and decision making has been studied intensively. (Elmachtoub and Grigas, 2022) introduced the SPO+ surrogate loss and proved consistency for problems with linear objectives, demonstrating that minimising prediction error alone can produce decisions arbitrarily far from optimal. (Donti et al., 2017) extended decision-focused learning to stochastic optimisation problems with differentiable solvers. (Wilder et al., 2019) treated combinatorial optimisation via a continuous relaxation of the optimiser. (Mandi et al., 2020) addressed hard combinatorial problems including knapsack and shortest-path. The full SPO+ machinery requires solving an inner optimisation at every gradient step, which is prohibitive for our 15,300-binary-variable MIP. We therefore adopt a tractable linear surrogate: an L1 loss weighted by the downstream sensitivity $\partial w / \partial f$ evaluated at a frozen reference allocation, which captures the leading-order regret gradient without per-batch inner solves.

For the uncertainty-handling components we follow standard references. (Birge and Louveaux, 2011) provide the canonical treatment of stochastic programming, including the VSS and EVPI definitions used in §3. (Soyster, 1973) and (Ben-Tal and Nemirovski, 2002) establish the robust optimisation framework used for the minimax formulation. Quantile regression (Koenker and Bassett Jr, 1978) supplies the scenarios. Demand prediction itself uses XGBoost (Chen and Guestrin, 2016) with SHAP attributions (Lundberg and Lee, 2017) for interpretability. The optimisation core is implemented in PuLP (Mitchell et al., 2011) and solved with CBC.

3 PROBLEM AND METHODOLOGY

3.1 Problem Description

PMPML’s network consists of three service classes: trunk routes (15–45 km, e.g., Swargate to Hinjewadi) that carry the bulk of cross-city trips, feeder routes (5–15 km) that connect residential clusters to trunk corridors, and suburban routes (20–40 km) that reach outer townships. Mean trip speed under typical Pune traffic is 17 km/h. Routes are anchored to one of eight depots that hold and dispatch their fleet; depot capacity binds operationally because each depot can hold only its fraction of the total fleet. Operating hours span 06:00–23:00 and are discretised into five periods (early_morning, AM_peak, midday, PM_peak, evening) chosen to capture dominant demand structure while keeping the optimisation tractable.

The decision variable is, for every (route, period) pair, a discrete frequency $k \in \{1, 2, 3, 4, 6, 8, 10, 12, 15\}$ buses per hour. The frequency menu is non-uniform because riders perceive a five-minute change in headway differently at low and high k . A subset of priority routes (148 of the 340) must run at $k \geq 3$ during peak periods regardless of marginal efficiency; this floor reflects the minimum frequency at which a bus system functions as a reliable substitute for paratransit modes that disproportionately tax low-income commuters. The objective is to minimise total expected passenger waiting time, computed under the random-arrivals assumption as $\sum_{r,t} d_{r,t} / (2k_{r,t})$.

3.2 Deterministic MIP Formulation

Let R be the set of routes ($|R| = 340$), T the set of operating periods ($|T| = 5$), and K the discrete frequency menu ($|K| = 9$). Parameters are demand $d_{r,t}$ (passengers per hour), cycle time τ_r (minutes), bus consumption $b_{r,k} = \lceil k\tau_r / 60 \rceil$, and the random-arrivals expected wait $w_k = 1 / (2k)$. The decision variable $x_{r,t,k} \in \{0, 1\}$ equals one if route r in period t is as-

signed frequency k :

$$\min_x \sum_{r,t,k} d_{r,t} w_k x_{r,t,k} \quad (1)$$

$$\text{s.t. } \sum_k x_{r,t,k} = 1 \quad \forall r, t \quad (2)$$

$$\sum_{r,k} b_{r,k} x_{r,t,k} \leq B \quad \forall t \quad (3)$$

$$\sum_{k \geq 3} x_{r,t,k} = 1 \quad \forall r \in R^P, t \in T^{\text{pk}} \quad (4)$$

$$\sum_{r \in R_j, k} b_{r,k} x_{r,t,k} \leq B_j \quad \forall j, t \quad (5)$$

Constraint (2) requires exactly one frequency per route-period. Constraint (3) caps the network-wide fleet at $B = 1800$. Constraint (4) is the equity floor for priority routes $r \in R^P$ during peak periods $T^{\text{pk}} = \{\text{AM_peak}, \text{PM_peak}\}$. Constraint (5) caps the buses anchored to depot j at the depot capacity B_j . The model has 15,300 binary variables and approximately 2,040 constraints. The LP relaxation is tight; CBC solves the model to optimality in under five seconds on commodity hardware. We adopt the standard random-arrivals expected-wait approximation $w_k = 1/(2k)$ following (Furth and Wilson, 1981); refinements that incorporate empirical headway variance are listed in §6.

3.3 Demand Forecasting

Data. Models are trained on hourly ridership for the 50 highest-demand routes over a single year, totalling approximately 430,000 rows after a seven-day lag-feature window is dropped. The remaining 290 routes are not modelled directly; their demand is scaled from the agency’s static demand matrix using residual ratios estimated on the modelled subset.

Features. Twenty-three features in total: cyclical encodings (sine and cosine) of hour, day-of-week, and month; calendar flags (weekend, holiday, monsoon, exam season, festival); weather (maximum temperature, precipitation in mm); route-level features (length, number of stops, priority score, one-hot service category); and lag features at one day and seven days at the same hour.

Models. Three model classes are compared. Linear regression provides the trivial baseline. Random forest (200 trees, default depth) provides a non-parametric tabular benchmark. XGBoost (Chen and Guestrin, 2016) with depth 8, learning rate 0.03, 2,000 trees with early stopping at patience 50, subsample 0.85, and column subsample 0.85 is the primary model. Three additional XGBoost models with quantile-regression objectives at $\tau \in \{0.1, 0.5, 0.9\}$

supply prediction intervals (Koenker and Bassett Jr, 1978).

Train/val/test split. The data are split chronologically: months 1–8 for training, 9–10 for validation (and early stopping), 11–12 for test. Random splits would leak future information into the training set and are not used.

3.4 Stochastic and Robust Optimisation

Scenarios. Six demand scenarios are constructed from the quantile predictions to span a realistic range of operational conditions. S1 (probability 0.35) is the baseline at $q_{0.5}$. S2 (0.15) is high demand at $q_{0.9}$. S3 (0.15) is low demand at $q_{0.1}$. S4 (0.15) is a monsoon shock dampening $q_{0.1}$ by an additional 20%. S5 (0.10) boosts trunk demand by 15% above $q_{0.9}$ during peak periods. S6 (0.10) is a festival dip scaling $q_{0.5}$ by 0.75 across the network. The scenario probabilities reflect a prior over the relative frequency of these conditions in a typical operating year.

Stochastic MIP. The stochastic formulation shares the deterministic decision variables but minimises the probability-weighted expected wait,

$$\min_x \sum_{s \in S} p_s \sum_{r,t,k} d_{r,t}^{(s)} w_k x_{r,t,k}, \quad (6)$$

which is computationally equivalent to the deterministic MIP with $d_{r,t}$ replaced by $\bar{d}_{r,t} = \sum_s p_s d_{r,t}^{(s)}$.

Robust MIP. The minimax formulation introduces an auxiliary scalar z and replaces the objective with $\min_x z$ subject to $z \geq \sum_{r,t,k} d_{r,t}^{(s)} w_k x_{r,t,k}$ for every scenario $s \in S$, plus all original constraints.

Information metrics. Following (Birge and Louveaux, 2011), $\text{VSS} = \text{EEV} - \text{RP}$ measures the value of the stochastic solution relative to the deterministic one, and $\text{EVPI} = \text{RP} - \text{WS}$ measures the value of perfect information about which scenario will materialise.

3.5 Decision-Focused Learning

The MSE-trained predictor weights all prediction errors equally, but the downstream optimisation does not: an error on a route-period whose optimal frequency is at the boundary between two discrete levels matters more than an error on a route-period whose optimum is robust to small perturbations. Because $\partial w / \partial f = -1/(2f^2)$, the optimisation sensitivity scales as $d_{r,t}/(2f_{r,t}^2)$ at the reference allocation $f_{r,t}$.

We test two decision-focused variants. The first is sample-weighted XGBoost: each training sample i on route r in period t is weighted by $w_i = d_{r,t}/(2f_{r,t}^2)$,

where $f_{r,t}$ is the optimal frequency from the deterministic MIP. The second is a two-layer fully-connected neural network (input $\rightarrow 128 \rightarrow 64 \rightarrow 1$, ReLU activations, BatchNorm) trained with the surrogate

$$L_{\text{SPO}^+}(\hat{d}, d) = \frac{1}{n} \sum_{i=1}^n \frac{d_i}{2f_i^2} |\hat{d}_i - d_i|. \quad (7)$$

The L1 form is robust to outliers, which matters because high-sensitivity samples dominate the loss surface and a single misbehaved point would otherwise destabilise the gradient. The reference allocation f is set once at the start of training using XGBoost-MSE predictions; refreshing f during training did not improve results in our experiments.

Connection to SPO+. The exact SPO+ loss (El-machtoub and Grigas, 2022) requires solving the inner MIP at every gradient step. Equation (7) is a linearisation that captures the leading-order sensitivity of the regret to prediction error at the fixed reference allocation while skipping the per-batch solve. The fixed-allocation approximation is defensible here because the optimal allocation structure is robust to small changes in demand: the binding constraints at the optimum are determined by the relative ordering of routes rather than by the absolute demand level.

4 EXPERIMENTS AND RESULTS

4.1 Experimental Setup

All MIPs are solved with PuLP (Mitchell et al., 2011) and CBC on a Windows 11 machine running Python 3.13.9, single thread for the solver, eight threads for XGBoost and random forest. Random seed 42 is used throughout. The full pipeline reproduces in approximately 75 minutes end-to-end. Code, data, models, and reproduction scripts will be released in a public repository upon acceptance.

4.2 Forecasting Results

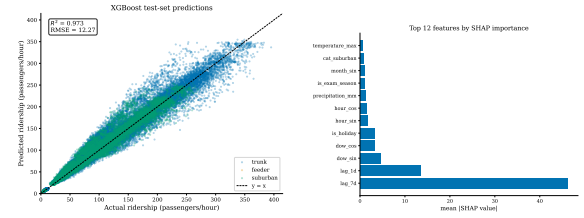
Table 1: Test-set forecasting performance (months 11–12).

Model	RMSE	MAE	MAPE	R ²
Linear regression	16.38	11.54	38.4%	0.952
Random forest	13.04	7.03	8.4%	0.970
XGBoost	12.27	6.96	8.8%	0.973

XGBoost achieves the best test-set performance on three of four metrics (Table 1). Random forest is competitive on MAPE, which over-weights small ridership values, but loses on RMSE, MAE, and R^2 . The

80% prediction interval covers 72.9% of test observations, slightly under-coverage relative to the nominal 80% but within the range of acceptable for an out-of-sample evaluation on a single year.

Figure 1a shows the joint distribution of actual and predicted ridership on the test set, coloured by route category. Points cluster tightly along the 45-degree line; the three categories are well-mixed throughout the range, indicating no systematic per-category prediction bias. Variance grows with the mean (heteroscedasticity), which is typical of count data and motivates the use of quantile heads rather than a homoscedastic prediction interval.



(a) Predictions vs. actuals. (b) SHAP feature importance.

Figure 1: Forecasting performance and feature importance.

Figure 1b ranks the top twelve features by mean absolute SHAP value. Lag features (one-day and seven-day) and the cyclical hour encoding dominate, consistent with the autoregressive structure of hourly ridership. Calendar flags and weather contribute at lower magnitudes. Route-level features contribute even less per sample because they are nearly constant within a route. Figure 2 shows the SHAP beeswarm: each point is one prediction; horizontal position is the signed contribution; colour encodes the feature value. Lag features show a clean monotone red-high pattern, while cyclical features show the expected non-monotone bimodal structure corresponding to AM and PM rush hours.

Figure 3 shows the time series for a representative route in each service category. The model captures the dominant weekly and daily structure. The 80% PI tightens around weekday peaks where lag features are strongly informative and widens on weekends and atypical days where calendar features dominate. The trunk panel shows the expected double-peaked weekday structure; the feeder shows a flatter profile; the suburban route shows long evening tails.

4.3 Optimisation Results

Table 2 reports five methods on a common constraint envelope. The MIP optimum reduces total waiting time by 26.0% relative to status quo while using es-

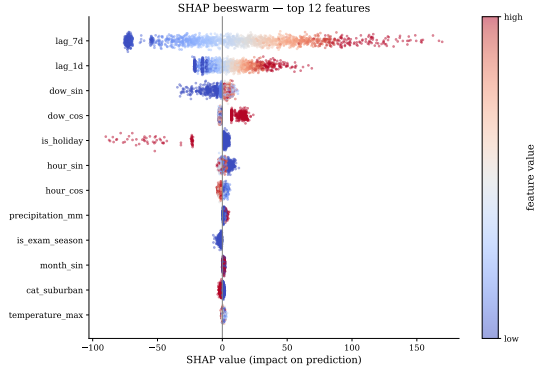


Figure 2: SHAP beeswarm for the top twelve features. Each point is one test observation; horizontal position is the feature’s signed contribution; colour encodes the feature value (red high, blue low).

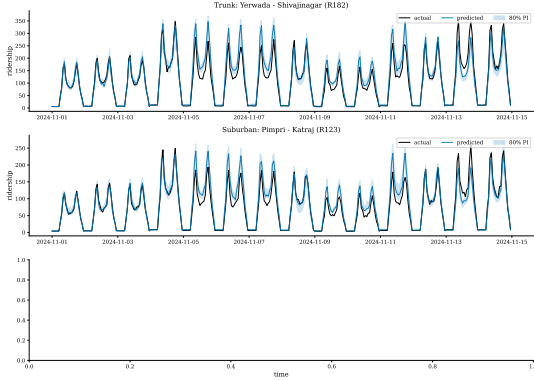


Figure 3: Two weeks of test data on a representative trunk, feeder, and suburban route. Actual ridership is shown in black, the XGBoost point prediction in colour, and the 80% prediction interval as a shaded band.

essentially the full fleet across all periods. Heuristic baselines underperform once forced through the same constraint set: the uniform policy cannot exceed $k = 2$ before depot caps bind; demand-proportional captures the right scaling but is locked into a one-parameter family; greedy upgrades route-periods by marginal benefit but lacks the lookahead to reach the global optimum.

Figure 4 compares the current PMPML allocation with the MIP optimum on the top thirty routes by total demand. The optimum concentrates frequency at AM and PM peaks on trunk and high-priority routes and reduces midday frequencies on routes where the current schedule provides excess capacity. Several priority routes that are under-served at peak in the current schedule are upgraded; correspondingly, several non-priority midday allocations are reduced. The pattern is consistent with the equity narrative: redistribution occurs primarily along the temporal axis rather than at the expense of priority routes.

Table 2: Method performance on the static demand matrix. Fleet used is summed across the five periods.

Method	Total wait	Fleet	% improv.
Status quo	22,748	7,300	–
Uniform	34,749	4,740	–52.8
Demand-proportional	21,046	7,917	7.5
Greedy	26,245	8,897	–15.4
Optimal MIP	16,826	9,000	26.0

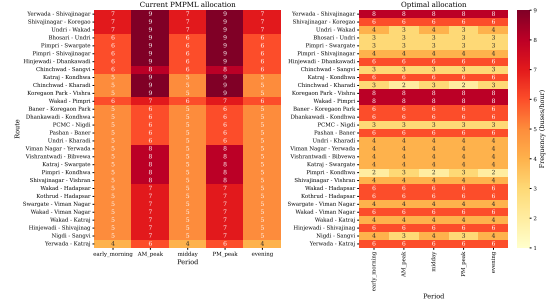


Figure 4: Top-30 routes by demand: current PMPML allocation (left) versus the MIP optimum (right). Frequency is in buses per hour; darker cells indicate higher frequency.

Figure 5 reports peak-frequency distributions split by priority status and by before/after. Three observations follow. Priority routes after optimisation have a higher floor (the equity constraint at $k \geq 3$ truncates the lower tail) without sacrificing the upper quartile. Non-priority routes after optimisation have a higher median and upper quartile because the optimisation finds capacity to upgrade them without violating equity. Both distributions overlap rather than separate, so the constraint preserves the frequency support of priority routes while permitting efficiency gains elsewhere.

4.4 Stochastic and Robust Analysis

Figure 6 reports the four optimisation methods (status quo, deterministic, stochastic, robust) under each of the six scenarios. The three optimised methods cluster tightly: each beats status quo by 20–30% in every scenario but they remain within a few hundred passenger-hours of each other. The robust solution achieves the lowest worst-case (scenario S5, the peak surge) at no detectable cost in expected performance. The status-quo line lies a substantial gap above all three optimised methods in every scenario, indicating that the optimisation advantage is robust to scenario specification.

VSS= 0.26 and EVPI= 2.47 passenger-hours (Figure 7). Both are positive but small. The small magnitude is informative: it indicates that the route-level structure of the optimal allocation is largely

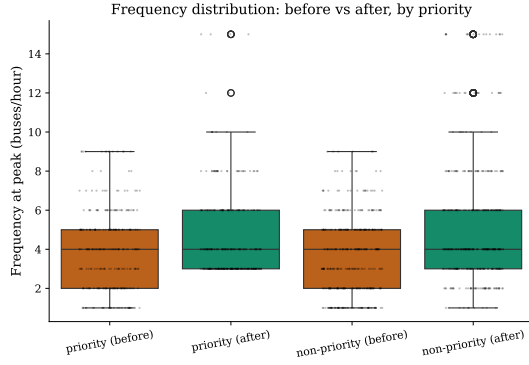


Figure 5: Frequency distribution at peak periods, before (current PMPML) and after (MIP optimum), split by priority status.

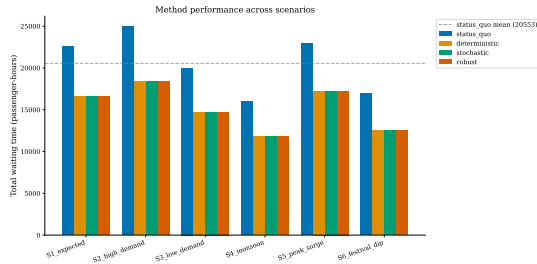


Figure 6: Method performance across the six demand scenarios. Bars are total waiting time. The dashed line is the status-quo mean across scenarios.

robust to scenario-level demand shifts, because the dominant source of variation is the relative ordering of routes (preserved across scenarios) rather than absolute demand levels. EVPI bounds the maximum value any further investment in scenario-resolution prediction could buy; at 2.47 passenger-hours, additional scenario modelling for this problem yields limited returns.

4.5 Decision-Focused Results

Table 3: Prediction quality versus decision quality. Regret is the method’s true objective minus the oracle’s true objective.

Method	RMSE	MAPE	Decision	Regret
XGBoost (MSE)	12.27	8.8%	16,713.04	0.130
XGBoost (weighted)	15.85	82.9%	16,713.37	0.459
NN (MSE)	12.75	13.6%	16,713.02	0.115
NN (SPO ⁺ surrogate)	13.94	35.2%	16,712.98	0.072
Oracle	—	—	16,712.91	0.000

Table 3 reports five methods evaluated on the same downstream task: predict demand, plug into the MIP, evaluate the resulting allocation against true test-period demand. The method with the lowest test

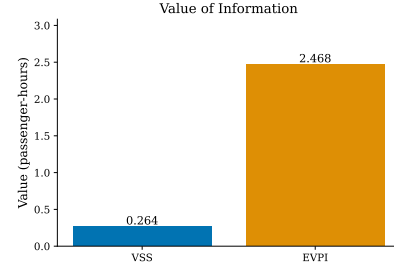


Figure 7: Value of Stochastic Solution (VSS) and Expected Value of Perfect Information (EVPI), in passenger-hours.

RMSE (XGBoost-MSE at 12.27) does not have the lowest regret (0.130); the method with the lowest regret (NN-SPO⁺ at 0.072) does not have the lowest RMSE (13.94). The SPO⁺ network achieves a 44.7% lower regret with a 13.5% larger RMSE.

The decision-weighted XGBoost variant fails as a forecasting tool: by weighting heavily on a few high-sensitivity samples it produces predictions that are systematically biased upward, with MAPE 82.9%. Its allocation is barely worse than the MSE baseline on regret (0.459 vs. 0.130), but the predictor itself is unusable in isolation. The NN-MSE matches XGBoost-MSE on regret (0.115 vs. 0.130) despite a slightly higher RMSE, suggesting that architectural flexibility helps somewhat even before decision-focused training is applied.

The economic interpretation: an agency selecting predictors by RMSE alone would deploy XGBoost-MSE and achieve regret 0.130; the same agency with the same data, using the SPO⁺ surrogate (7), would achieve regret 0.072 from identical training data. The 44.7% reduction requires no additional data, features, or compute – only a different loss function.

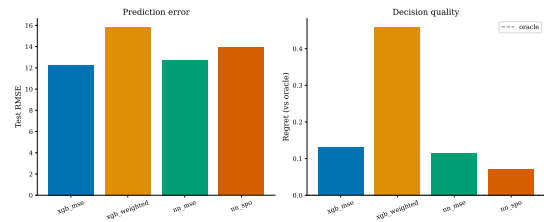


Figure 8: Prediction error (left) and decision regret (right) across four predict-then-optimize methods. The dashed line indicates the oracle reference. The method with smallest RMSE is not the method with smallest regret.

Figure 8 visualises the dissociation. The bar with the smallest RMSE (XGBoost-MSE) is not the bar with the smallest regret (NN-SPO⁺); the cross-panel comparison makes this point visible that a single-axis plot would obscure.

4.6 Sensitivity Analysis

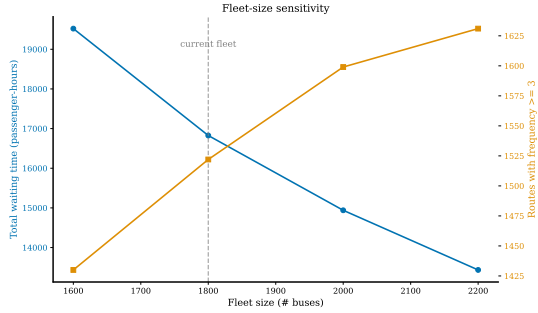


Figure 9: Fleet-size sensitivity. Left axis: total waiting time. Right axis: number of route-periods with frequency at least three. Vertical dashed line: current fleet $B = 1800$.

Figure 9 sweeps fleet size from $B = 800$ to $B = 2200$. Below $B = 1600$ the model is infeasible because depot capacity scales with fleet size and the equity floor cannot be met simultaneously across all depots at low fleet. From $B = 1600$ upward the curve is monotone decreasing, with diminishing returns: each additional 200 buses removes 1,500–2,000 passenger-hours of waiting time. The right-axis count of route-periods with $k \geq 3$ rises from 1,430 at $B = 1600$ to 1,631 at $B = 2200$, quantifying how many additional route-periods can be lifted above the “functional service” threshold as fleet grows.

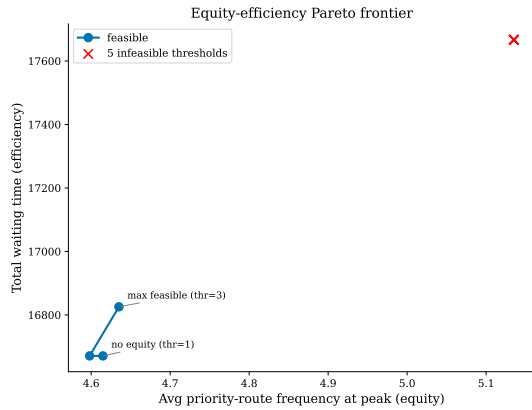


Figure 10: Equity-efficiency Pareto frontier. Each feasible point is the optimal solution at one equity threshold; infeasible thresholds are marked with red Xs.

Figure 10 sweeps the equity threshold from one (no constraint) to ten (every priority route runs at least ten buses per hour at peak). Thresholds 1–3 are feasible at $B = 1800$. Threshold 4 is the cliff: the feasible region collapses because depots D5 and D6, which together hold 125 of the 148 priority routes, cannot simultaneously support a four-bus-per-hour floor and

the rest of the network at any positive frequency. Total waiting time rises from 16,671 (threshold 1) to 16,826 (threshold 3), an increase of 0.9%. The current equity floor costs less than one percent of total waiting time relative to a counterfactual with no equity constraint.

The LP-relaxation shadow prices identify the depot bottlenecks quantitatively: D5 at AM peak has marginal value -5.56 passenger-hours per additional bus capacity, and D6 at AM peak has marginal value -9.40 . These are the binding bottlenecks that, if relaxed, would yield the largest reduction in total waiting time per bus of additional capacity.

5 DISCUSSION

Operational implications. The headline result is that the same fleet, redistributed according to the optimisation, yields a 26.0% reduction in total passenger waiting time. At PMPML’s approximately 800,000 daily riders the absolute recovery is on the order of 1,000 passenger-hours per day. The implementation requirement is procedural rather than capital: the agency would need to update its frequency tables and verify that the new schedule is feasible at each depot. The shadow-price analysis (§4.6) provides specific guidance on which depots bind operationally.

Decision-focused dissociation. The empirical finding that the predictor with the worst RMSE achieves the best regret (§4.5) has direct methodological implications. Standard ML benchmarks (RMSE, MAPE, R^2) are weak proxies for operational decision quality in predict-then-optimize pipelines, and the gap is large enough to be operationally significant. The fix is essentially free: identical data, architecture, and compute, with a loss function that incorporates downstream sensitivity (7), yield a 44.7% regret reduction. The lesson generalises to other operational settings (inventory, scheduling, capacity planning) where a predictor feeds an optimiser.

Equity as a near-free policy choice. The Pareto frontier indicates that the current equity floor of three buses per hour costs only 0.9% of total waiting time relative to a counterfactual with no floor. The cliff at threshold four is operational rather than welfare-driven: it reflects the binding of depot capacity, not a fundamental incompatibility between equity and efficiency. The policy implication is that agencies should preserve the current floor and target the next round of capital investment at the binding depots specifically.

Generalisability. The framework ports directly to other Indian transit agencies with similar institutional structures: BEST (Mumbai), BMTC (Bangalore), DTC (Delhi). The components that require

agency-specific re-fitting are route metadata, demand data, and depot capacities. The optimisation core and the decision-focused training procedure are agency-agnostic.

6 LIMITATIONS AND FUTURE WORK

Synthetic ridership for unmodelled routes. The demand model is trained on 50 routes; the remaining 290 are assigned demand by scaling the static demand matrix using residual ratios estimated on the modelled subset. A fully data-driven analysis would require smart-card or AFC data on every route. The agency’s data-sharing policy is the binding constraint, not the methodology.

Static daily allocation. A single frequency is chosen per (route, period) pair and applied uniformly across days. Real PMPML operations adjust schedules for day-of-week and special events. The framework can in principle handle these by re-optimising at any cadence; the static analysis is presented because the operational and equity questions resolve at this level.

Random-arrivals waiting model. The expected wait $w_k = 1/(2k)$ assumes Poisson arrivals and uniform headways. Real bus arrivals exhibit bunching (Adamski’s bus paradox) that makes the effective wait longer than $1/(2k)$, especially on heavily-loaded routes. Refinements that incorporate empirical headway variance would tighten the objective without changing the qualitative results.

Demand under price shocks. The demand model treats fares and competition as exogenous. In reality, paratransit operators reprice in response to bus service quality. A genuinely closed-loop extension would treat demand as a function of frequency through modal substitution.

Decision-focused approximation. The SPO^+ surrogate (7) is a linearisation around a fixed reference allocation. Refresh strategies (e.g., updating the reference every E epochs) and benchmarks against the full SPO^+ loss with an inner solver on a coarsened MIP are deferred to future work.

Single-year training window. The forecast model is trained on a single year of data. Multi-year training would improve estimates of festival and seasonal effects, although the operational allocation is dominated by within-week and within-day structure that the single-year window already captures.

Hard equity floor. The current formulation treats equity as a hard constraint at $k \geq 3$ for priority routes during peaks. Soft penalties, fairness-aware objec-

tives (Wilder et al., 2019), and geographic equity formulations are alternative specifications that may be politically or practically preferable in some settings.

Future directions. (i) Integration of real smart-card data once available; (ii) real-time re-optimisation in response to disruptions, leveraging the sub-five-second solve time; (iii) multi-modal optimisation that couples frequency with modal-substitution behaviour; (iv) joint frequency-pricing optimisation as practised by some international agencies; (v) decision-focused architecture search beyond the two variants tested here, including graph neural networks on the route topology and proper SPO^+ losses with inner solvers on coarsened MIPs; (vi) richer equity formalisations from the algorithmic-fairness literature; (vii) cross-agency benchmarking on BEST, BMTC, and DTC.

7 CONCLUSION

This paper presented a complete predict-then-optimize pipeline for the bus frequency allocation problem of Pune’s PMPML network. The pipeline integrates a deterministic MIP that respects fleet, depot, and equity constraints, an XGBoost demand model with quantile heads, stochastic and robust formulations driven by quantile-derived scenarios, and a decision-focused neural network trained with an SPO^+ -style surrogate.

Three findings stand out. First, the optimisation reduces total waiting time by 26.0% at the same fleet size relative to current PMPML frequencies, with depot bottlenecks at D5 and D6 identified by LP shadow-price analysis. Second, the decision-focused neural network achieves a 44.7% lower decision regret than the MSE-trained XGBoost despite a 13.5% larger RMSE, demonstrating that prediction quality and decision quality dissociate cleanly on this problem and that the dissociation is operationally significant. Third, the current equity floor of three buses per hour at peak for priority routes costs 0.9% of total waiting time, with the feasibility cliff at threshold four arising from depot capacity rather than from a fundamental equity-efficiency conflict.

The methodological contribution – that prediction quality and decision quality must be evaluated separately for predict-then-optimize pipelines – generalises beyond transit. The practical contribution is a turnkey pipeline that reproduces in approximately 75 minutes and is portable to other Indian transit agencies. Code, data, and reproduction scripts will be released upon acceptance.

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